

SIMATS ENGINEERING

ASSESSMENT TOOL 2 — INDUSTRY PROBLEM SOLVING SOLUTIONS

COURSE NAME	Data Warehousing and Data Mining	COURSE CODE	CSA16
WEIGHTAGE	40%	TOTAL MARKS	35 (5 Questions × 7 Marks)

COURSE OUTCOME COVERED (CO4): Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

Industry Problem Solving Solutions & Architectural Designs

Q1. Supermarket Purchases Customer Segmentation (BL3 – 7 Marks)

Given Dataset:

Customer	Visits/Month	Avg Spend (₹)	Items Bought
C1	10	2,000	15
C2	4	5,000	8
C3	12	1,800	18
C4	3	6,000	6
C5	11	2,100	16

CLUSTERING APPROACH & MARKETING STRATEGY

1. Proposed Clustering Approach:
- **Algorithm Selection: K-Means Clustering** (or K-Medoids) with $K = 2$ clusters following Standard/Min-Max Feature Normalization.
 - **Feature Analysis & Cluster Formation:**
 - **Cluster 1 ("High-Frequency Budget Shoppers"):** {C1, C3, C5} — High visits (~11/month), lower spend per visit (~₹1,967), high items count (~16 items).
 - **Cluster 2 ("Low-Frequency Premium Shoppers"):** {C2, C4} — Low visits (~3.5/month), high spend per visit (~₹5,500), low items count (~7 items).
2. Support for Targeted Marketing Strategies:
- **For High-Frequency Shoppers ({C1, C3, C5}):** Drive basket size expansion via volume discounts, loyalty points per visit, and cross-category multi-buy promotions.
 - **For Premium Shoppers ({C2, C4}):** Focus on retention and visit frequency increase using targeted premium product catalogs, exclusive weekend sales invitations, and high-margin product bundles.

Q2. Credit Card Transaction Fraud Detection (BL4 – 7 Marks)

Given Dataset:

Transaction	Amount (₹)	Location	Time (hrs)
T1	200	Chennai	10
T2	5,000	Delhi	2
T3	150	Chennai	11

Transaction	Amount (₹)	Location	Time (hrs)
T4	7,000	Mumbai	1
T5	180	Chennai	12

CLUSTERING-BASED FRAUD DETECTION SYSTEM DESIGN

1. Clustering-Based Anomaly Detection Design:
- Selected Algorithm: DBSCAN (Density-Based Spatial Clustering of Applications with Noise) or Isolation Forest / Distance-to-Medoid Thresholding.
 - Feature Encoding & Distance Metric: One-hot encoding for Location, continuous features for Amount and Time (02:00 = 2 hrs, 01:00 = 1 hr).
 - Cluster Separation Results:
 - Dense Core Cluster (Legitimate Routine Transactions): {T1, T3, T5} — Local transactions in Chennai during regular daytime hours (10:00–12:00) with small amounts (₹150–₹200).
 - Outliers / Anomalies (Flagged Fraud): {T2, T4} — High-value transactions (₹5,000–₹7,000) occurring at non-standard night hours (01:00–02:00) from non-primary locations (Delhi, Mumbai).
2. Industry Workflow & Actionable Interventions:
- Real-time Outlier Scoring: Transactions with high distance to core dense clusters trigger an instant hold.
 - Automated Fraud Controls: Requires immediate SMS/App Two-Factor Authentication (2FA) or temporary card block for transactions T2 and T4.

Q3. Telecom User Usage & Churn Prevention (BL4 – 7 Marks)

Given Dataset:

User	Call Duration (min)	Data Usage (GB)	Recharge (₹)
U1	300	5.0	200
U2	100	2.0	100
U3	350	6.0	220
U4	80	1.0	90
U5	320	5.5	210

CHURN IDENTIFICATION & CUSTOMER RETENTION STRATEGY

1. Clustering Application for Churn Profiling:
- Methodology: Apply K-Means Clustering (K=2) or Agglomerative Clustering on scaled usage features.
 - Customer Segments Identified:
 - Cluster A ("Active / High-Value Users"): {U1, U3, U5} — Call duration >300 min, data usage 5–6 GB, recharge ₹200–₹220. High network engagement and low churn risk.
 - Cluster B ("Low Engagement / Churn-Prone Users"): {U2, U4} — Call duration <100 min, data usage ≤2 GB, recharge ≤₹100. Disengaging activity pattern indicates high churn probability.
2. Targeted Retention Strategies:
- Proactive Win-Back Offers for Cluster B ({U2, U4}): Deploy personalized discount recharges, bonus data add-ons, or bundled streaming subscriptions to reactivate engagement.
 - Service Health Surveys: Send automated feedback inquiries to identify network coverage issues or dissatisfaction before complete subscriber loss.

Q4. E-Commerce User Behavior & Recommendation Systems (BL3 – 7 Marks)

Given Dataset:

User	Product Views	Clicks	Purchases
U1	50	20	5
U2	30	10	2
U3	60	25	6
U4	20	8	1
U5	55	22	5

CLUSTERING TECHNIQUE & RECOMMENDATION ENGINE INTEGRATION

1. Suggested Clustering Technique:

- **Technique:** Hierarchical Agglomerative Clustering (HAC) or K-Means ($K=2$) integrated with Collaborative Filtering.
- **User Groupings Identified:**
 - **Group 1 ("High-Intent Power Buyers"):** {U1, U3, U5} — Views 50–60, Clicks 20–25, Purchases 5–6. Conversion rate ~10%.
 - **Group 2 ("Casual / Low-Activity Browsers"):** {U2, U4} — Views 20–30, Clicks 8–10, Purchases 1–2.

2. Recommendation System Enhancements:

- **Cluster-Based Collaborative Filtering (CBCF):** Recommends items purchased by other members of the same cluster, vastly reducing the sparsity problem of raw user-item matrices.
- **Dynamic UI/UX Personalization:**
 - *For Power Buyers ({U1, U3, U5}):* Show instant "1-Click Checkout" and premium personalized cross-sells.
 - *For Casual Browsers ({U2, U4}):* Display trending deals, customer reviews, and limited-time coupons to lower conversion friction.

Q5. Traffic Data Congestion Zone Identification (BL4 – 7 Marks)

Given Dataset:

Location	Vehicle Count	Avg Speed (km/h)	Delay (min)
L1	200	30	10
L2	500	15	25
L3	220	28	12
L4	600	10	30
L5	210	32	9

TRAFFIC MANAGEMENT & CONGESTION IDENTIFICATION SOLUTION

1. Clustering-Based Congestion Zone Detection:

- **Clustering Model:** K-Means Clustering ($K=2$) or Density-Based Clustering on standardized traffic metrics.
- **Traffic Zone Classification:**
 - **Zone 1 ("Smooth Flow / Non-Congested"):** {L1, L3, L5} — Low vehicle count (200–220), high speed (28–32 km/h), minimal delay (9–12 min).
 - **Zone 2 ("Bottleneck / Severe Congestion"):** {L2, L4} — High vehicle count (500–600), low speed (10–15 km/h), high delay (25–30 min).

2. Operational Traffic Management Applications:

- **Adaptive Traffic Signal Control:** Automatically extend green-light durations for roads entering Zone 2 locations (L2, L4).
- **Dynamic Navigation & Re-routing:** Feed real-time cluster status into GPS systems (e.g., Google Maps) to divert incoming traffic away from L2/L4 to alternative corridors.
- **Infrastructure Planning:** Prioritize flyover/bypass construction and traffic police deployment at identified chronic congestion clusters.